

4. SYSTEM ANALYSIS AND CLASSIFICATION

Objective

This experiment is divided into four simulations:

1. Discrete-Time System
2. Causal/Non-Causal System
3. Analysis of Discrete System
4. Analysis of Continuous-Time System

Experiment 4.1

Discrete -Time System

Program:

```
// Test Case: Analysis of a Discrete-Time Linear System

// Define parameters
N = 20; // Number of samples
n = 0:N-1; // Discrete time index vector
x_n = ones(1, N); // Input sequence: Unit step sequence
y_n = zeros(1, N); // Output sequence initialization
// Define the difference equation:  $y[n] = 0.5y[n-1] + x[n]$ 
for k = 2:N
    y_n(k) = 0.5 * y_n(k-1) + x_n(k);
end

// Plot the input sequence
clf; // Clear the current figure
subplot(2,1,1);
plot2d (n, x_n);
xlabel('n (Discrete Time Index)');
ylabel('Amplitude');
title('Input Sequence: Unit Step Sequence');

// Plot the output sequence
subplot(2,1,2);
```

```

plot2d(n, y_n);

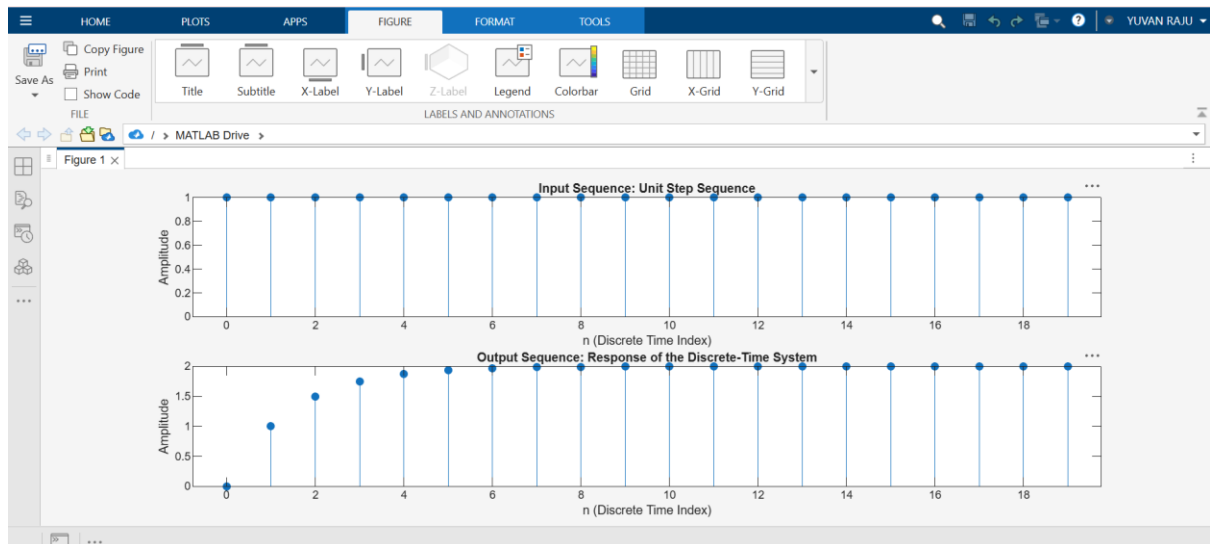
xlabel('n (Discrete Time Index)');

ylabel('Amplitude');

title('Output Sequence: Response of the Discrete-Time System');

```

OUTPUT



Experiment 4.2

Causal/Non-Causal System

Program:

```

// Test Case: Causal and Non-Causal Systems Analysis

// Define parameters

N = 20; // Number of samples

n = 0:N-1; // Discrete time index vector

x_n = ones(1, N); // Input sequence: Unit step sequence

// Initialize output sequences

y_causal = zeros(1, N); // Output sequence for causal system

y_noncausal = zeros(1, N); // Output sequence for non-causal system

// Causal System:  $y[n] = x[n] + 0.5x[n-1]$ 

for k = 1:N

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if k > 1 then
y_causal(k) = x_n(k) + 0.5 * x_n(k-1);
else
y_causal(k) = x_n(k); // No x[n-1] for k=0
end
end

// Non-Causal System:  $y[n] = 0.5x[n] + 0.5x[n+1]$ 
for k = 1:N
if k < N then
y_noncausal(k) = 0.5 * x_n(k) + 0.5 * x_n(k+1);
else
y_noncausal(k) = 0.5 * x_n(k); // No x[n+1] for k=N-1
end
end

// Plot the input sequence
clf; // Clear the current figure
subplot(3,1,1);
plot2d(n, x_n);
xlabel('n (Discrete Time Index)');
ylabel('Amplitude');
title('Input Sequence: Unit Step Sequence');

// Plot the output of the causal system
subplot(3,1,2);
plot2d(n, y_causal);
xlabel('n (Discrete Time Index)');
ylabel('Amplitude');
title('Output Sequence: Causal System');

// Plot the output of the non-causal system
subplot(3,1,3);
plot2d(n, y_noncausal);

```

```
xlabel('n (Discrete Time Index)');  
  
ylabel('Amplitude');  
  
title('Output Sequence: Non-Causal System');
```

OUTPUT

